3.5.1 Contamination of Food

Food contamination occurs by substances (contaminants) not intentionally added to food. Contaminants are the factors responsible for unhygienic food. They compromise food safety and cause harm to the health of a consumer. Such substances may be chemical, physical or biological.

Chemical

(For example, chemical poisons like insecticide) Chemical poisons such as insecticides get into food, and toxic metals may enter food during processing. Poisonous plants (and fungi) like some types of mushrooms and seafood produce chemicals or toxins which can cause illnesses if consumed incorrectly.

Physical

(For example, undesirable substances in food) Reports of 'foreign bodies' such as dead rats, insects and pieces of glass in food get wide publicity although they are rare events. Physical contaminants such as these are usually detected by the consumer and the food is not consumed. However, substances like glass or staple pins used for packing can be dangerous. These incidents rarely cause food poisoning but are, of course, highly undesirable.

Biological

(For example, bacteria, their toxins and viruses) Biological contaminants include microorganisms/microbes which are small organisms that can only be seen through a microscope. The most common types of microorganisms are bacteria and viruses.

3.6 Food borne Diseases

Food-borne diseases, including food-borne intoxications and food-borne infections, are terms applied to illnesses acquired through consumption of contaminated food, and are also frequently referred to as food poisoning.

While many food-borne diseases may be selflimiting, some can be very serious and can lead to death particularly in children, pregnant women and older persons.

3.6.1 Classification of food-borne illnesses

- **Food-borne infections** – caused by consuming foods or liquids contaminated with bacteria, viruses, or parasites. These pathogens cause infection by:
  - Invading and multiplying in the lining of the intestines and/or other tissues
  - Invading and multiplying in the intestinal tract and releasing a toxin (bacteria only).
- **Food-borne intoxications** – caused by consuming foods or beverages already contaminated with a toxin. Sources of toxins are as follows:
  - Certain bacteria (pre-formed toxins)
  - Poisonous chemicals
  - Natural toxins found in animals, plants, and fungi.

Salmonella food Poisoning:

Causative agents: Salmonella group are the main group of bacterial agents

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